

Predicting α -helix secondary structure

A two-state Hidden Markov Model fit with amortized BayesFlow inference

Group [number]

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Roadmap

1

The question Predict α -helix vs other from sequence

2

Forward–Backward Exact per-residue posterior as training targets

3

BayesFlow Amortized neural posterior via a sliding window

4

Insulin & real evaluation Held-out real proteins vs ground truth

5

Comparison & results BayesFlow vs exact Forward–Backward

From amino-acid sequence to α -helix

- Proteins fold; the local shape is the secondary structure (helix / sheet / coil).
- We predict one thing: is each residue an α -helix, or 'other'?
- Solving 3-D structure needs X-ray / NMR — expensive; we want it from sequence.
- Our statistical model: a two-state Hidden Markov Model.

α -helix = state “H”

“other” = E B G I T S C

Mapping decision

DSSP labels each residue with 8 states. We keep only H as helix (strict α -helix); the other 7 $\rightarrow$ other.

The generative model: a two-state HMM

- The chain always starts in “other”.
- Transitions: helix→helix 0.90, other→helix 0.05.
- So helices form ~10-residue runs, and are started rarely.
- Realistic: real data gives 0.912 and 0.041 (see next slide).

Prior & likelihood

No parameters are inferred — all fixed. The prior is the induced distribution over hidden state paths $p(z_{1:T})$, from start = [0,1] and the transition matrix. The likelihood is the emission table.

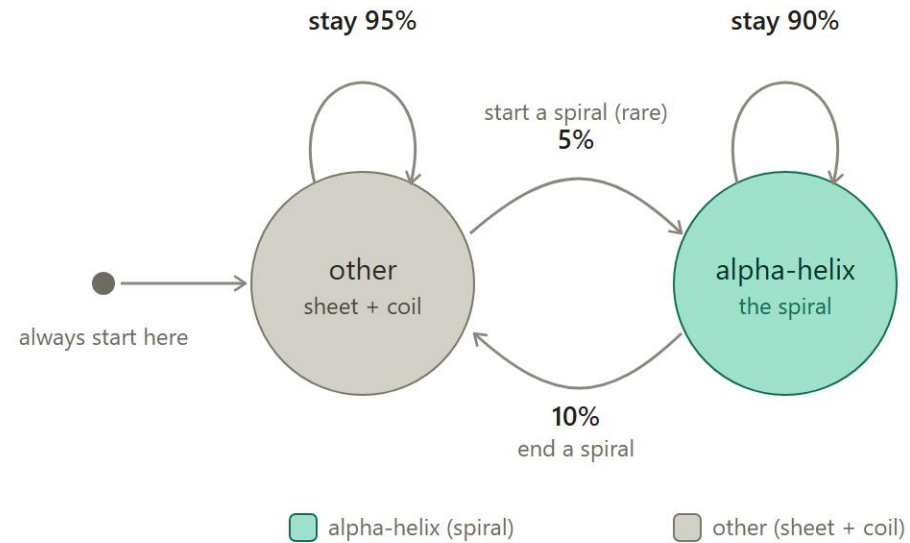


Fig 1. Two-state HMM: start rule and transition probabilities.

Emissions — and are they realistic?

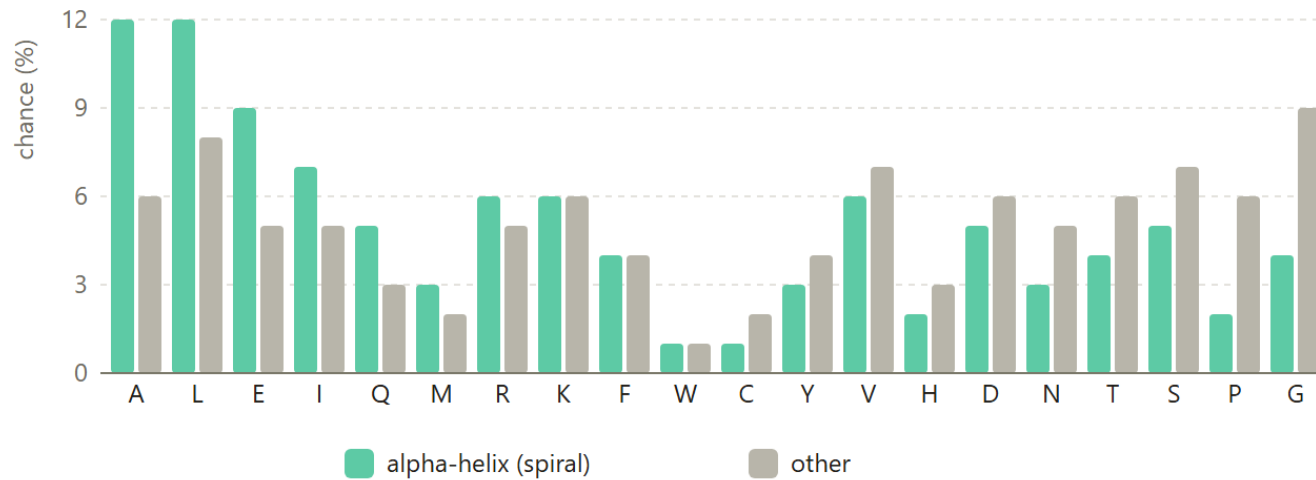


Fig 2. Per-state amino-acid emission probabilities (α -helix vs other).

Yes — checked against real PISCES data: emissions match at $r = 0.99$ (max 0.9 pp off); transitions match too ($P(h \rightarrow h)$ 0.912 vs 0.90). Both tables are empirically well-founded.

Notebooks 1–2: simulate, then get exact targets

- Simulate 100,000 chains; lengths sampled from real PISCES (median 208).
- Run Forward–Backward (hmmlearn) → exact $P(\text{helix})$ per residue = training targets.
- 24.3M residues — the exact Bayesian “answer key”.
- Checks: mean P 0.324; residue-0 = 0; recovers hidden states at the Bayes ceiling (AUC 0.79).

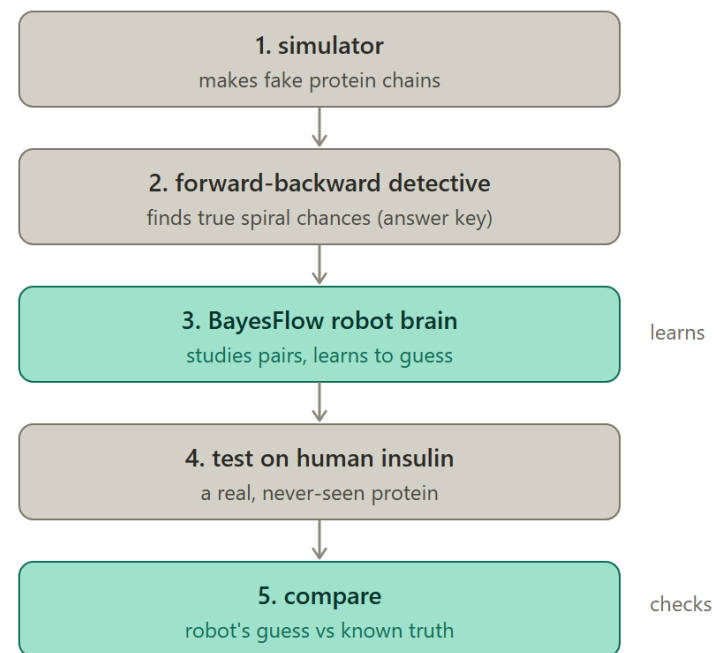


Fig 3. Full pipeline: simulator → Forward–Backward → BayesFlow → evaluation.

Turning a variable-length chain into a fixed input

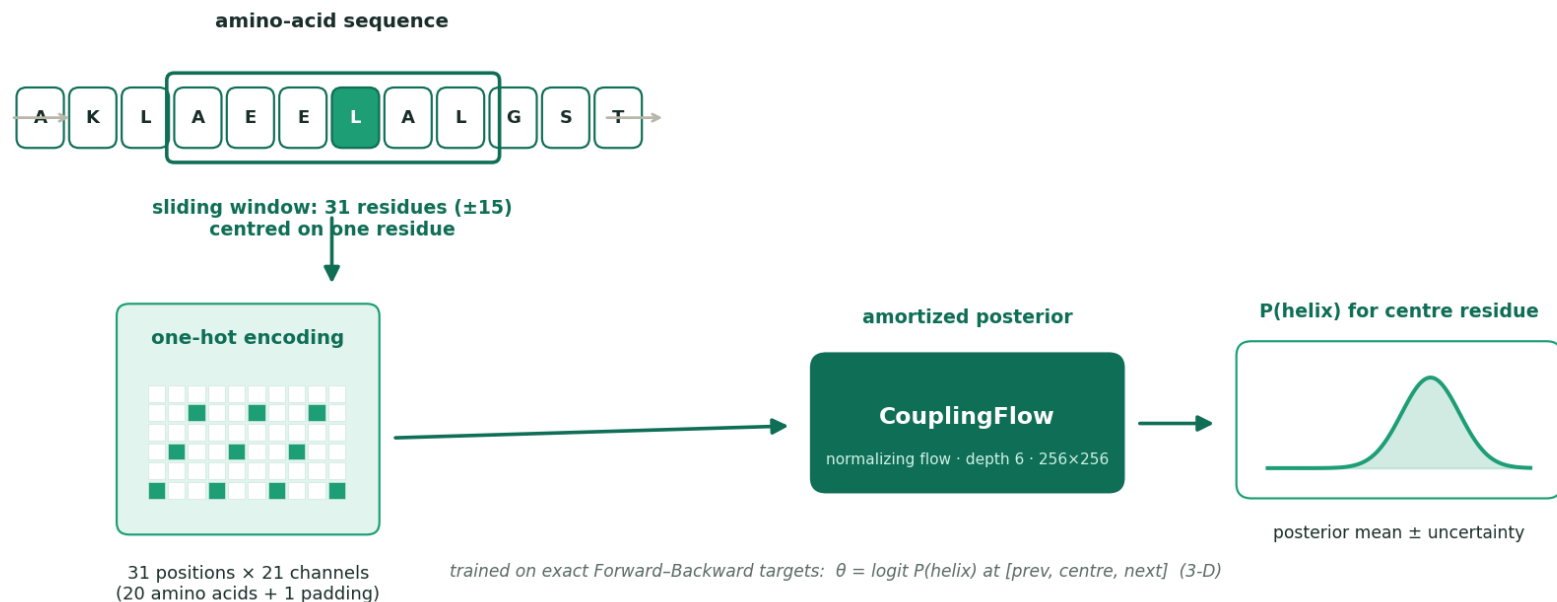


Fig 4. Sliding 31-residue window $\rightarrow$ one-hot encoding $\rightarrow$ CouplingFlow posterior.

- Flow needs a fixed-size input $\rightarrow$ predict one residue from a fixed 31-residue window (± 15), slid along the chain.
- One-hot: 31 $\times$ 21 channels = 20 amino acids + 1 padding (chain ends).
- Given the FULL sequence, γ is DETERMINISTIC (a point mass) — nothing to estimate. We condition on a LOCAL window so γ becomes uncertain $\rightarrow$ a genuine posterior to learn.

Why ± 15 is enough

2nd eigenvalue $\lambda_2 = 1 - 0.10 - 0.05 = 0.85 \rightarrow$ influence decays geometrically: $0.85^{15} \approx 0.09$. Residues outside the window carry $< 10\%$. So $r = 0.999$ is expected, not suspicious.

How we trained it

Component	Choice · why
Inference network	CouplingFlow, depth 6, 256×256 subnets
Target θ (3-D)	logit $P(\text{helix})$ at [prev, centre, next]
Conditions	flattened $31 \times 21 = 651$ one-hot window
Training data	300,000 windows from 15,000 chains
Optimizer / epochs	Adam · 30 epochs · batch 512

No leakage (verified)

Train = front block [0, 15000); validation = tail block [99500, 100000). Disjoint indices AND 0 shared sequences — asserted at runtime.

Why SBI if the likelihood is tractable?

Because tractability is the point. The exact posterior is KNOWN here, so we can verify the amortized estimator is faithful — validation that is impossible in a real SBI problem.

Held-out real protein: wild-type insulin (1MSO)

- Insulin + true DSSP labels from the dataset; model trained ONLY on simulated data (unseen).
- B-chain: AUC 0.98 [95% CI 0.93–1.00] — robust, excludes chance. Its helix is propensity-driven (L,V,E,A).
- A-chain: AUC 0.53 [95% CI 0.25–0.79] — CI straddles chance, so $n=21$ residues can't settle it.
- Hypothesis consistent with the data: the A-chain helix is disulfide-stabilised & Cys-rich, and C disfavors helix in our table — a sequence-only HMM cannot see 3-D bonds.

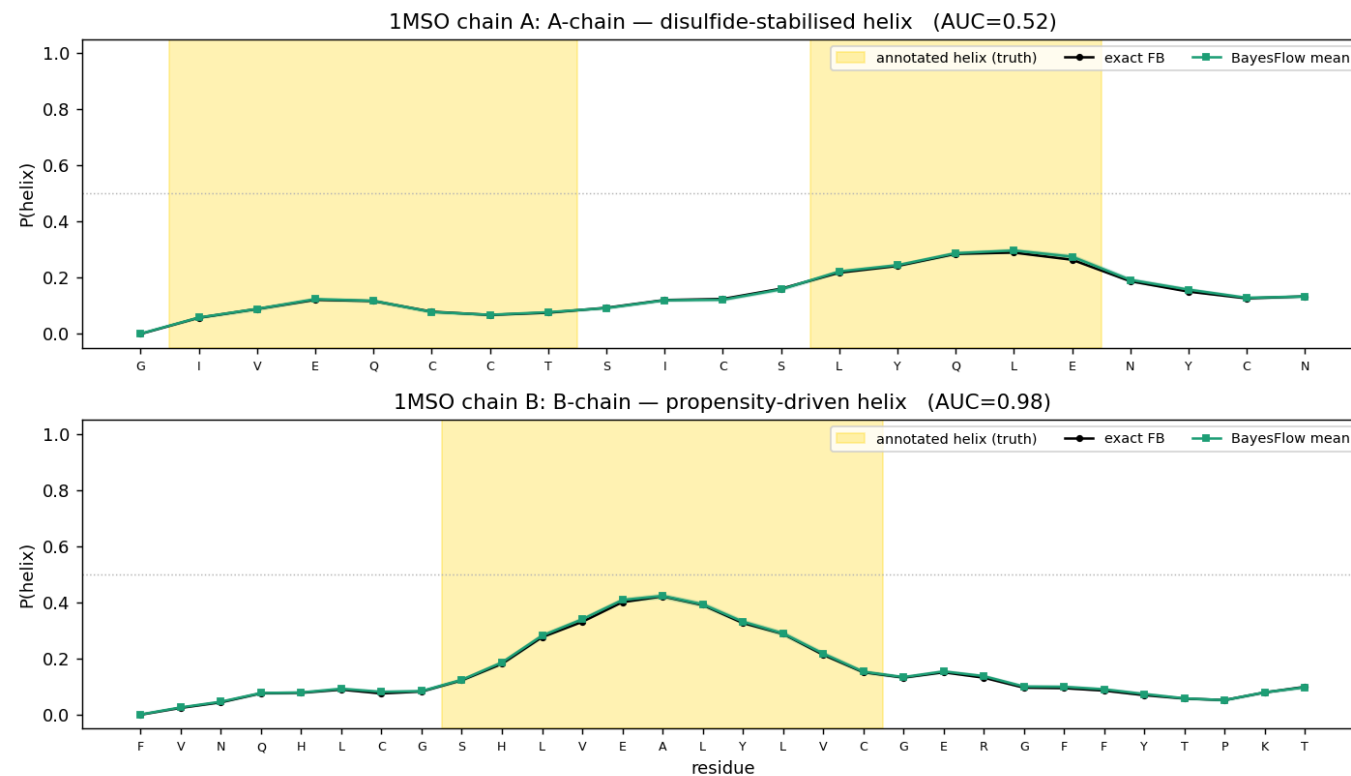


Fig 5. Insulin $P(\text{helix})$: BayesFlow (green) vs exact FB (black); gold = true helix.

Real-data headline: all PISCES proteins

- This — not insulin — is the real-data evidence: 8,308 chains, 2.2M residues.
- Per-chain AUC 0.753, 95% CI [0.750, 0.756]; 96.7% of chains beat chance.
- BayesFlow = exact FB (0.754); a simple sequence-only HMM ranks helices well.
- AUC > accuracy: the HMM isn't calibrated to reality → use ranking.

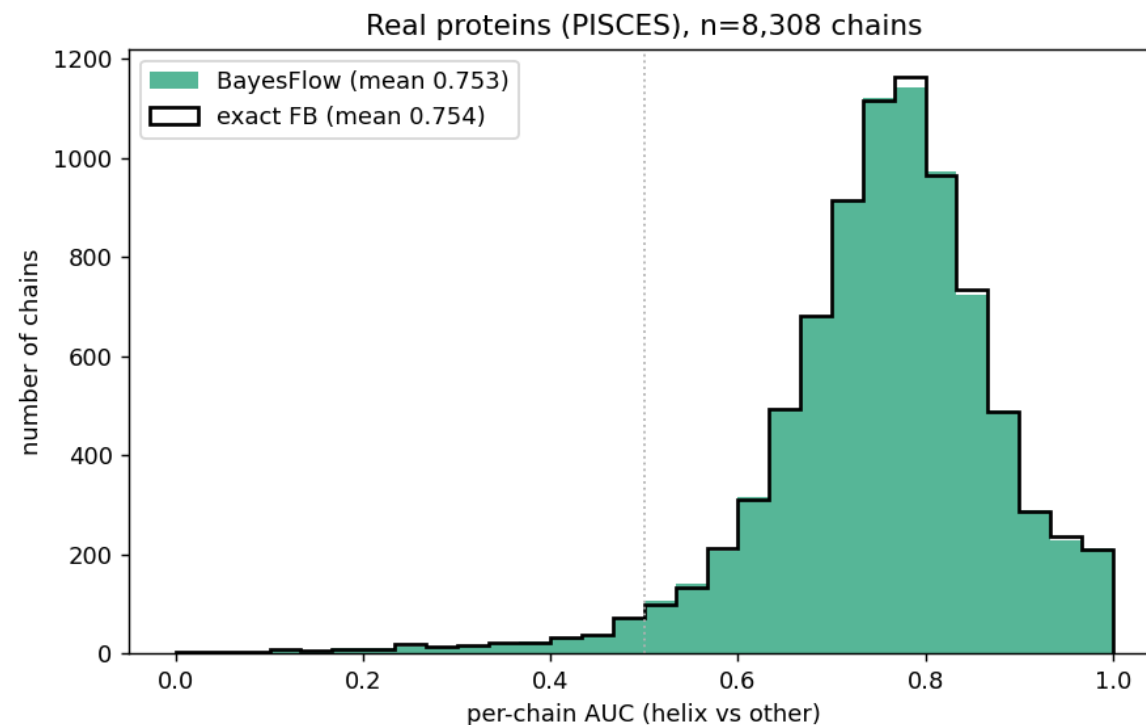


Fig 6. Per-chain AUC over 8,308 real chains — the quantitative real-data claim.

Result 1: the neural posterior reproduces exact inference

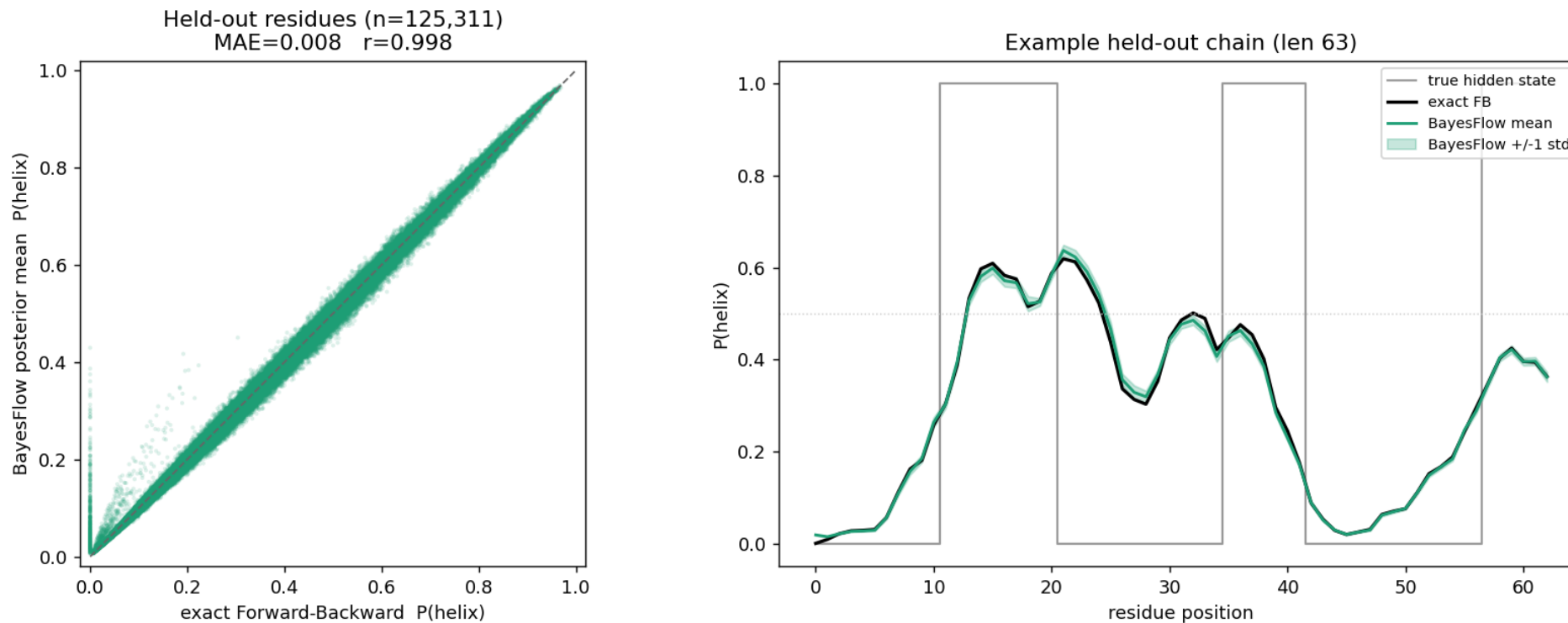


Fig 7. Left: BayesFlow mean vs exact FB ($r=0.999$). Right: an example chain with the ± 1 std band.

Held-out simulated chains: correlation 0.999, MAE 0.007. Expected, not suspicious: $\lambda_2^{15} \approx 0.09$, so the ± 15 window already carries nearly all the information.

Calibration (SBC): we don't just report it — we control it

Calibration is controlled by window size (same FlowMatching model)

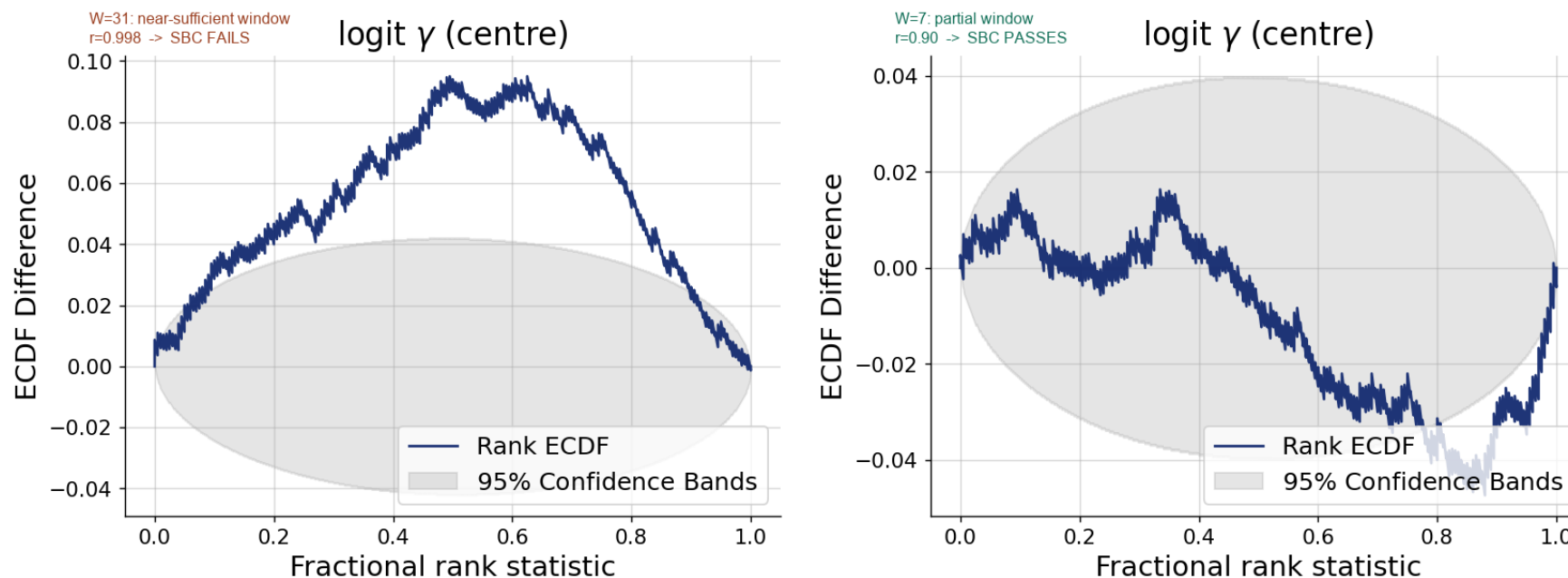


Fig 8. SBC rank ECDF (same FlowMatching model): shrinking the window restores genuine uncertainty $\rightarrow$ SBC passes.

- Convergence + recovery ($r=0.998$) + contraction (0.998) all excellent. SBC on the headline ($W=31$) model FAILS: a tiny ~ 0.15 -SD bias (≈ 0.002 in probability).
- Diagnosed: the window carries $>90\%$ of the info ($\lambda_2^{15} \approx 0.09$), so γ | window is near-deterministic — ANY flow struggles (two architectures fail at the same magnitude).
- Demonstrated control: shrink the window ($W=31 \rightarrow 7$) $\rightarrow$ genuine uncertainty $\rightarrow$ SBC PASSES, trading point accuracy (r 0.998 $\rightarrow$ 0.90). A tunable accuracy $\leftrightarrow$ calibration trade-off, not a bug.

New finding: the accuracy $\leftrightarrow$ calibration frontier

Accuracy $\leftrightarrow$ calibration trade-off across window size (FlowMatching model)

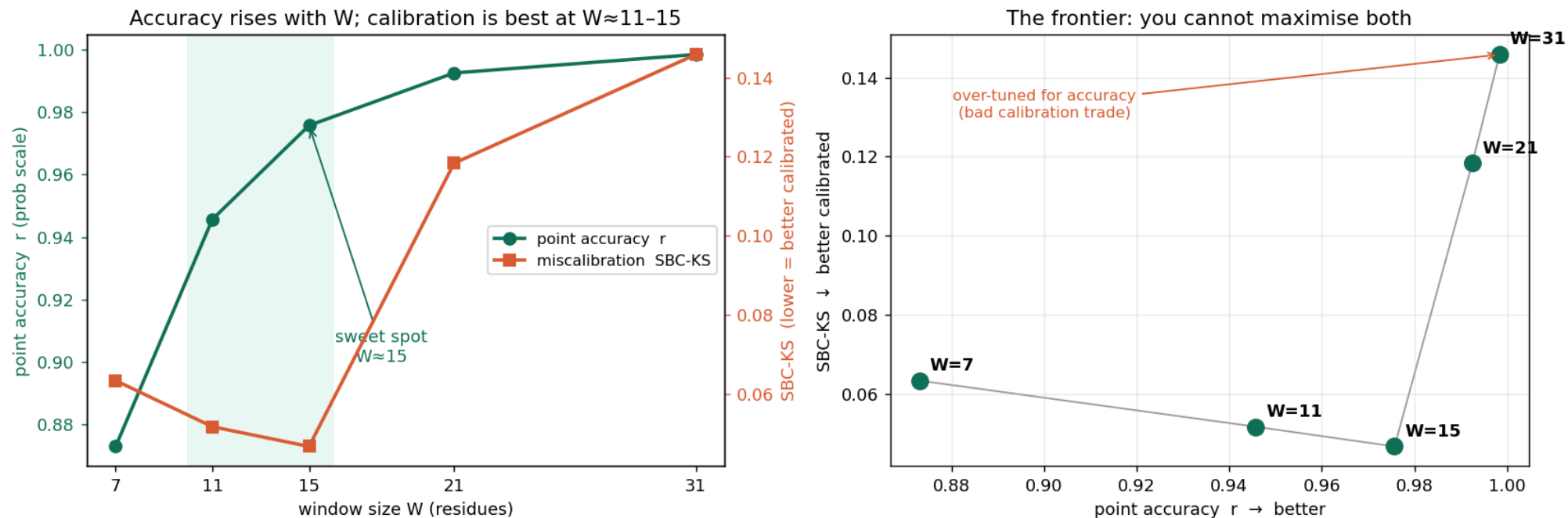


Fig 9. Window-size sweep: point accuracy (r) and miscalibration (SBC-KS) for $W = 7 \dots 31$.

- We swept the window over 5 sizes and measured BOTH point accuracy (r) and calibration (SBC-KS) on held-out chains.
- Accuracy rises monotonically with W ; calibration is best at $W \approx 11-15$, then degrades sharply (2–3 $\times$) beyond it.
- So the headline $W=31$ model is OVER-TUNED for accuracy: +0.02 r costs +0.10 KS. $W \approx 15$ is near-Pareto-optimal ($r=0.976$ with the best calibration) — the trade-off is a design knob, not a bug.

Results at a glance

Setting	AUC BayesFlow	AUC exact FB	Acc@0.5 BayesFlow	Acc@0.5 exact FB	Majority baseline
Held-out simulated	0.797	0.798	0.760	0.760	0.674
Insulin B (wild-type 1MSO)	0.981	0.986	0.633	0.633	0.633
Insulin A (wild-type 1MSO)	0.519	0.509	0.429	0.429	0.571
Real PISCES (8,308 chains)	0.753	0.754	0.728	0.728	0.708

Table 1. AUC & accuracy@0.5 vs the majority-class baseline — BayesFlow vs exact FB.

- BayesFlow $\approx$ exact FB in every setting — faithful amortization.
- Accuracy@0.5 barely clears the majority-class baseline on real data (0.728 vs 0.708) and ties/loses on insulin $\rightarrow$ AUC is the fair metric, not accuracy.
- FB is a ceiling on SIMULATED data only; on real proteins the HMM is misspecified, so a better model can beat it.

TL;DR — take-home message

- α -helix as a 2-state HMM whose transition AND emission tables match real proteins ($r \approx 0.99$).
- The likelihood is tractable by design: because the exact posterior is known, we can verify the amortized estimator is faithful — it reproduces exact FB at $r = 0.999$ (validation impossible in a real SBI problem).
- Diagnostics: SBC reveals a calibration bias on the headline model — but a window-size sweep shows it is a TUNABLE accuracy $\leftrightarrow$ calibration trade-off (best at $W \approx 15$), not a bug.
- On unseen real proteins it ranks propensity-driven helices well (insulin B-chain 0.98 [0.93–1.00]; 8,994 PISCES chains ≈ 0.75) — the honest limit of a sequence-only model.

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